

Unit- 3

Number System

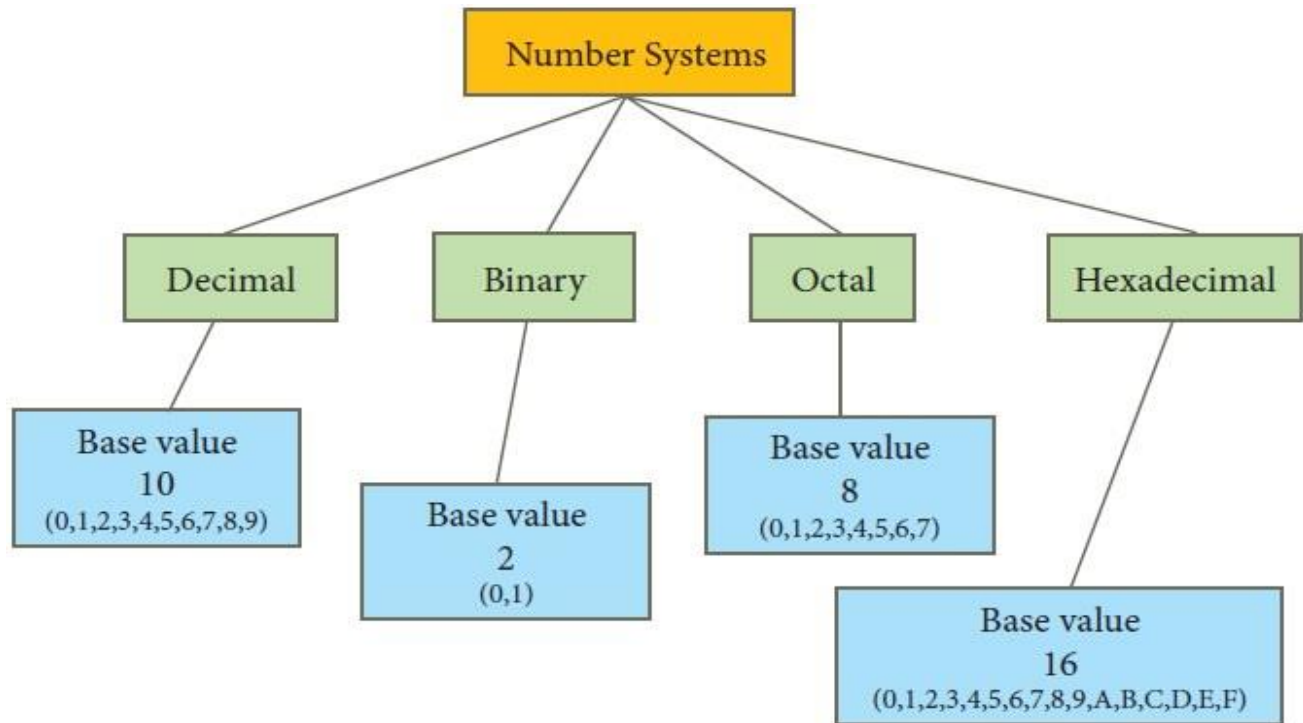


Figure 2.3. Number Systems

★ Convert Binary Number to Decimal Number :-

➤ Decimal Number Base=10

➤ Binary Number Base=2

➤ **Formula :- $N \dots 16, 8, 4, 2, 1$**

➤ Ex :- $(10011)_2$

16 8 4 2 1

1 0 0 1 1

$=16+2+1=(19)_{10}$

2. $(111001)_2 = 32+16+8+1$

$=(57)_{10}$

32 16 8 4 2 1

1 1 1 0 0 1

3. $(110010)_2$

$32+16+2=(50)_{10}$

❖ Convert Decimal Number To Binary Number :-

Ex.....(19)₁₀

$$= 16+2+1 \text{ (10011)}_2$$

$$16 \ 8 \ 4 \ 2 \ 1$$

$$(1 \ 0 \ 0 \ 1 \ 1)$$

1. (43)₁₀

$$= 32+8+2+1 = (101011)_2$$

2. (97)₁₀

$$= 63+32+1 = (1100001)_2$$

3. (47)₁₀

$$= 32+8+4+2+1 = (101111)_2$$

4. (65)₁₀

$$= 64+1 = (1000001)_2$$

5. (86)₁₀

$$= 64+16+4+2 = (1010110)_2$$

6. (98)₁₀

$$= 64+32+2 = (1100010)_2$$

7. (102)₁₀

$$= 64+32+4+2 = (100110)_2$$

8. (13)₁₀

$$= 8+4+1 = (1101)_2$$

❖ Convert Octal To Decimal Numbers :-

Ex... (645)₈

$$= 6*8^2 + 4*8^1 + 5*8^0$$

$$= 6*64 + 32 + 5$$

$$= (421)_{10}$$

1. (378)₈

$$= 3*8^2 + 7*8^1 + 8*8^0$$

$$= 3*64 + 56 + 8$$

$$= 192 + 56 + 8$$

$$= (256)_{10}$$

2. (370)₈

$$= 3*8^2 + 7*8^1 + 0*8^0$$

$$= 192 + 56$$

$$= (248)_{10}$$

3. (540)₈

$$= 5*8^2 + 4*8^1 + 0*8^0$$

$$= 320 + 32 + 0$$

$$= (352)_{10}$$

4. (420)₈

$$= 4*8^2 + 2*8^1 + 0*8^0$$

$$= 256 + 16 + 0$$

$$= (272)_{10}$$

5. (680)₈

$$= 6*8^2 + 8*8^1 + 0*8^0$$

$$= 384 + 64 + 0$$

$$= (448)_{10}$$

❖ Convert Decimal To Octal Numbers :-

Ex...(80)₁₀ =(120)₈

Example 1: Convert (127)₁₀ to Octal.

Solution: Divide 127 by 8

$127 \div 8 = 15$ (Quotient) and (7)Remainder

Divide 15 by 8 again.

$15 \div 8 = 1$ (Quotient) and (7) Remainder

Divide 1 by 8, we get;

$1 \div 8 = 0$ (Quotient) and (1) Remainder

Since the quotient is zero now, no more division can be done. So by taking the remainders in reverse order, we get the equivalent octal number.

Hence, $(127)_{10} = (177)_8$

Example 2: Convert 52₁₀ to octal.

Solution: Divide 52 by 8

$52 \div 8 = 6$ (Quotient) and (4)Remainder

Divide 6 by 8 again.

$6 \div 8 = 0$ (Quotient) and (6) Remainder

Since the quotient is zero now, no more division can be done. So by taking the remainders in reverse order, we get the equivalent octal number.

Hence, $(52)_{10} = (64)_8$

Example 3: Convert 100₁₀ to octal.

Solution: Divide 100 by 8

$100 \div 8 = 12$ (Quotient) and (4)Remainder

Divide 12 by 8 again.

$12 \div 8 = 1$ (Quotient) and (4) Remainder

Divide 1 by 8, we get;

$1 \div 8 = 0$ (Quotient) and (1) Remainder

Since the quotient is zero now, no more division can be done. So by taking the remainders in reverse order, we get the equivalent octal number.

Hence, $(100)_{10} = (144)_8$

❖ Convert Octal To Binary Number :-

➤ Binary-2

➤ Octal-8 $2^3=8$

Octal Number	Equivalent Binary Number
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

1. $(154)_8$

$= (001101100)_2$

❖ **Convert Binary To Octal Number :-**

1. $(110,101)_2$

$= (65)_8$

2. $(111,001)_2$

$= (71)_8$

3. $(100110)_2$

$= (46)_8$

❖ **Convert Hexa Decimal To Decimal :-** (0 TO 15)

0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

A	1	0	1	0
B	1	0	1	1
C	1	1	0	0
D	1	1	0	1
E	1	1	1	0
F	1	1	1	1

1. $(37B)_{16} = (?)_{10}$
 $= (3 \cdot 16^2) + (7 \cdot 16^1) + (11 \cdot 16^0)$
 $= 768 + 112 + 11$
 $(891)_{10}$
2. $(3B)_{16} = (?)_{10}$
 $= (3 \cdot 16) + (11 \cdot 16^0)$
 $= 48 + 11$
 $(59)_{10}$
3. $(E7A9)_{16} = (?)_{10}$
 $= (14 \cdot 16^3) + (7 \cdot 16^2) + (10 \cdot 16^1) + (9 \cdot 16^0)$
 $= (4096 \cdot 14) + (7 \cdot 256) + (160) + (9)$
 $= 57344 + 1792 + 160 + 9$
 $(59305)_{10}$

❖ **Convert Hexadecimal TO Binary :-**

1. $(122D)_{16} = (?)_2$
 $= 1 \quad 2 \quad 2 \quad D$
 $0001 \quad 0010 \quad 0010 \quad 1101$
 $= (0001001000101101)_2$
2. $(56A5D)_{16} = (?)_2$
 $= 5 \quad 6 \quad A \quad 5 \quad D$
 $0101 \quad 0110 \quad 1010 \quad 0101 \quad 1101$
 $= (01010110101001011101)_2$
3. $(20E)_{16} = (?)_2$
 $= 2 \quad 0 \quad E$
 $0010 \quad .0000 \quad 1110$
 $= (001000001110)_2$

❖ **Convert Binary of Hexadecimal :-**

1. $(110110111)_2 = (?)_{16}$
 $= \underline{1} \quad \underline{1011} \quad \underline{0111}$

1 B 7

$= (1B7)_{16}$

2. $(111101011101)_2 = (?)_{16}$

$= \underline{1111} \quad \underline{0101} \quad \underline{1101}$

F 5 D

$= (F5D)_{16}$

3. $(10101111)_2 = (?)_{16}$

$= \underline{1010} \quad \underline{1111}$

A F

$(AF)_{16}$